

A test stand for the High-Luminosity LHC

The major upgrade of the LHC will be tested in an above-ground facility that has recently had its first superconducting system installed



Two fully synchronised overhead cranes were used to handle the cold powering system, which is made up of a long transmission line and an impressive connection system. (Image: CERN)

An impressive operation recently took place in CERN's magnet test hall. The innovative cold powering system has been successfully installed in the [High-Luminosity LHC](#) (HL-LHC) Inner Triplet (IT) String test stand. This novel system comprises a long electrical transmission line, which has been specially developed to transport currents to the magnets across a wide range of temperatures. Its installation in the IT String follows on from the installation of the novel protection system and is an important milestone in the development of the HL-LHC.

The High Luminosity LHC (HL-LHC) is a major upgrade of CERN's [Large Hadron Collider](#) (LHC), which aims to increase the number of particle collisions (luminosity) and consequently boost the amount of physics data that can be collected, allowing further discoveries to be made.

Innovative beam-focusing magnets, known as inner triplets, are a major part of this upgrade. These magnets will be deployed on both sides of the beam interaction points at the [ATLAS](#) and [CMS](#) experiments with new powering, protection and alignment systems and – just like the LHC magnets – they will operate at 1.9 K (an extremely cold temperature, colder than deep outer space).

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Karel Šafařík (1953 – 2024)

CERN is currently constructing a test string for the HL-LHC in the above-ground LHC test hall. It will be composed of the six main superconducting beam-focusing magnets – the inner triplets – and the associated technology, and the layout will reproduce the underground configuration of the LHC. After each system is individually validated, the IT String will be used to validate the integration of the full range of systems.

“The project will test superconducting magnet circuits under conditions as close as possible to those they will experience in the HL-LHC tunnel. The primary goal is to enable teams to optimise the installation of these components, plan for potential repair work or interventions in the tunnel and study the collective behaviour of major components”, explains Marta Bajko, head of the IT String team.

The [cold powering system](#) transfers the current from the power converters to the magnets and is composed of a approximately 75 m-long high-temperature-based superconducting link made of novel superconducting materials (e.g. magnesium diboride). We say it is “high-temperature-based” because of its extraordinary ability to transport 120 kA of current, in a compact volume, from the 20 power converters – which sit in a new HL-LHC tunnel built specially for them and operate at room temperature – down to the magnets (which are kept at the extremely cold temperature of 1.9 K) in the LHC tunnel, with almost no energy loss.

“This milestone follows about ten years of development on different aspects of the cold powering system. Eight of these cold powering systems will be installed underground in the LHC after full qualification. The CERN transport team has been instrumental in this complex installation

process,” says Amalia Ballarino, leader of the HL-LHC cold powering system.



Adjusting the superconducting link as it is unspooled into the IT String. (Image: CERN)

The handling of the cold powering system, weighing about 5 tonnes, required two fully synchronised overhead cranes and a whole team to manually move and adjust its position as it was wound onto, and then off of, a huge spool.

“Before carrying out the manoeuvre, we developed a complex integration and assembly procedure, conducting a risk analysis at each step. This involved meticulous studies and simulations and extensive real-life testing campaigns by the entire team”, explains Stefanos Spathopoulos, the CERN engineer who was in charge of the design and production of key mechanical components and the planning of the operation.

These activities are paving the way for the next stages of the IT String, with the next major step being the installation of the magnets.

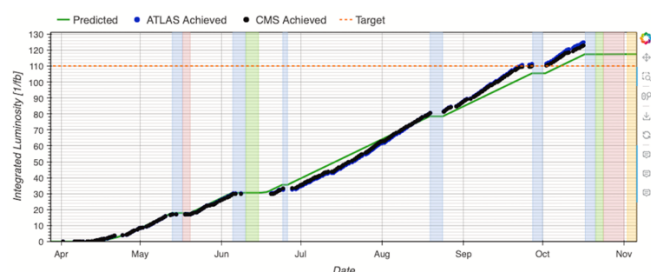
Florence Thompson

Accelerator Report: ending the 2024 LHC proton run on a high note

This year’s LHC proton run ended on 16 October, surpassing expectations with 11% more collisions than planned. Next up: the lead-ion run

This year’s LHC proton run, which has seen protons colliding at an energy of 6.8 TeV, ended on 16 October, surpassing expectations with 11% more collisions than planned. Next up: a four-day machine development (MD) block, before the LHC

gears up for its final run of the year, the lead-ion run, scheduled to end at 6.00 a.m. on 25 November.



The integrated luminosity evolution throughout the 2024 proton run for ATLAS and CMS. The green line shows the predicted integrated luminosity, while the orange dotted line is the updated target of 110 fb^{-1} . The black and blue dots show the integrated luminosity achieved by CMS and ATLAS respectively. (Image: CERN)

The end of the proton run and the transition to the lead-ion run is a nice moment to briefly look back at what we have achieved so far. This year's proton run began on 11 March with four weeks of beam commissioning, during which the LHC was successfully set up using a new "reverse polarity" optics scheme. The aim was to reduce the effects of radiation on key magnets near the ATLAS experiment (see [the 14 March Accelerator Report](#)). By 5 April, the LHC had already achieved stable beams and collisions with a small number of proton bunches, with experiments [starting to collect data three days ahead of schedule](#).

The intensity ramped up very smoothly to about 1800 bunches but, on 17 April, unexpected beam losses occurred, indicating that the collimator hierarchy was broken (see [the 3 May Accelerator Report](#)).

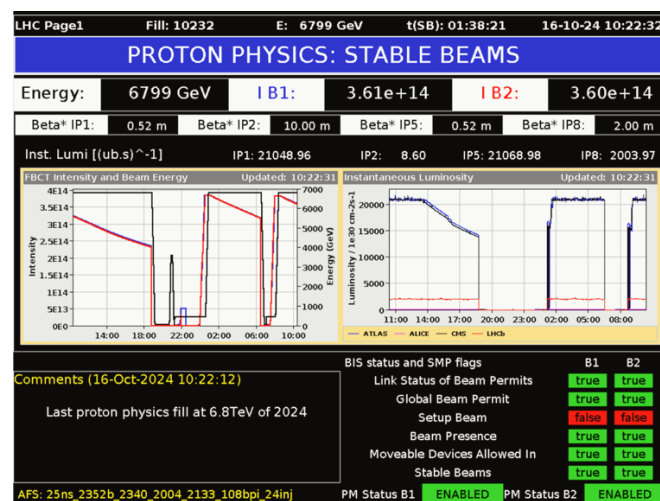
The machine performance was limited while extensive studies and tests followed, which led to the development of a successful solution that was implemented and validated after the scheduled technical stop from 10 to 15 June. Nevertheless, the intensity ramp-up was completed on 29 April, when we managed to inject 2352 bunches per beam.

In the meantime, the LHC injectors switched to another beam production scheme, providing the BCMS (Beam Compression, Merging and Splitting) beam – a brighter beam designed to deliver more collisions (see [the 30 May Accelerator Report](#)). Additionally, the LHC Machine Committee approved a four-week extension of the 2024 run on 24 April, raising the luminosity target from 90 fb^{-1} to 110 fb^{-1} .

After the technical stop, the restart of luminosity production was challenging and progress initially

lagged behind predictions. However, by mid-August, the pace picked up and the LHC exceeded expectations, delivering a final integrated luminosity of 124 fb^{-1} for both ATLAS and CMS – about 11% higher than the 110 fb^{-1} target. LHCb and ALICE also surpassed their goals, achieving 10.6 fb^{-1} and 61.6 pb^{-1} respectively.

From the start of physics operations on 25 April to the conclusion of the proton run on 16 October, the LHC was operational 73% of the time and in collision mode 52% of the time. This remarkable performance wouldn't have been possible without the outstanding availability, performance and flexibility of the injector complex, which continuously adapted to the LHC's needs and pushed the machines' performance whenever possible.



The last proton physics fill at 6.8 TeV of the year took place on 16 October 2024. (Image: CERN)

After the four-day LHC MD block, a three-day technical stop will take place to prepare for the upcoming lead-ion run. Interestingly, the lead-ion run isn't just about colliding lead ions – it also includes a brief period of proton–proton collisions at 2.68 TeV. This serves as a sort of calibration for the lead-ion collisions.

Once the proton and lead-ion beams are set up, there will be six days of proton–proton collisions at 2.68 TeV, followed by 19 days of lead–lead collisions. However, these lead-ion collisions present significant challenges. During the 2023 run, the LHC machine experienced beam losses during acceleration that prematurely dumped the beams, and the ALICE experiment observed high background noise, complicating data collection (see [the 12 October 2023 Accelerator Report](#)). To

address these issues, extensive studies and tests were conducted and several improvements were implemented, resulting in record-breaking data collection for ALICE.

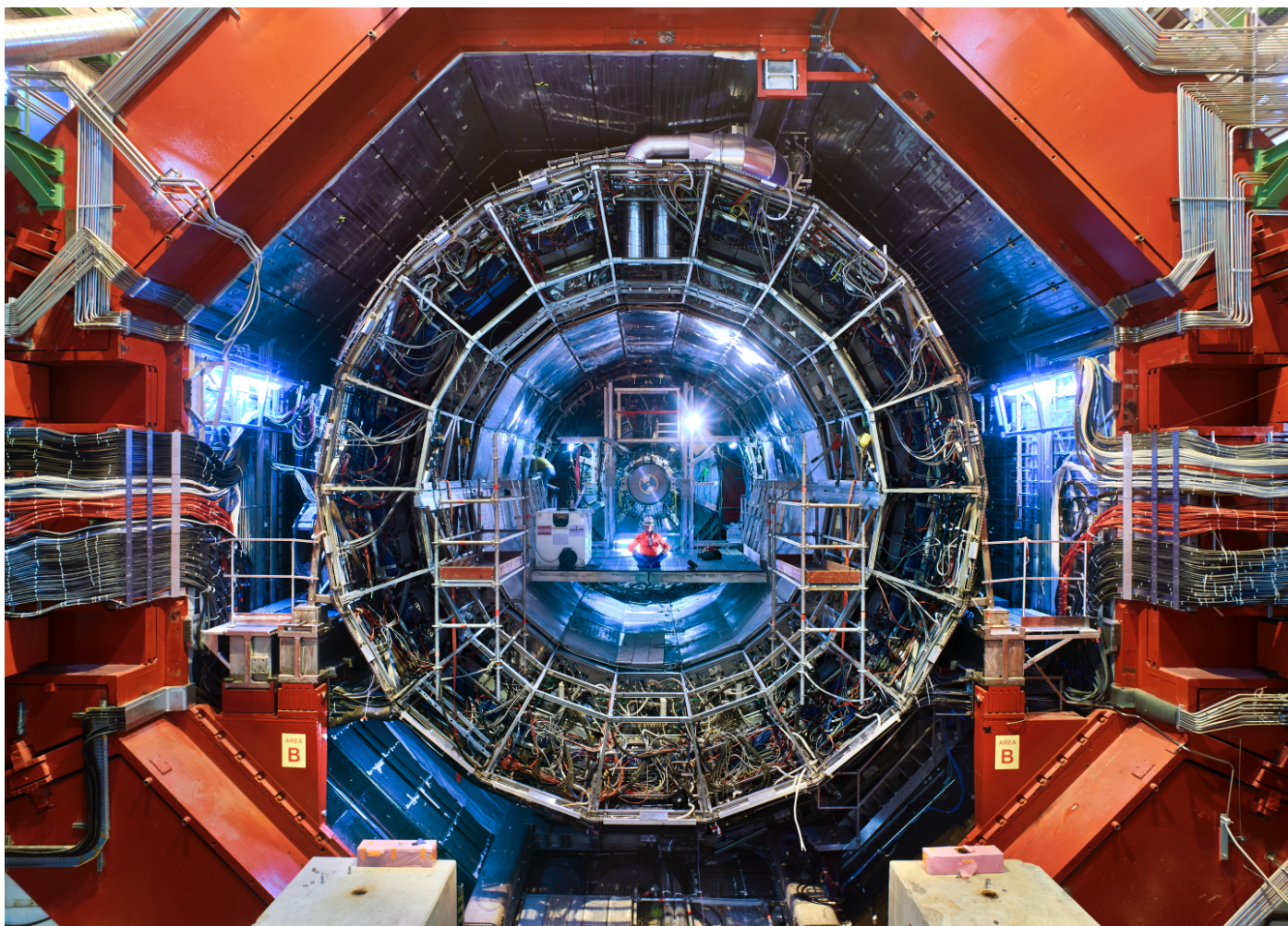
Still, there is a need for further enhancements in beam performance and background reduction to

ensure a smooth and successful 2024 lead-ion run. Stay tuned!

Rende Steerenberg

CERN70: Tasting the primordial soup

Part 19 of the [CERN70](#) feature series. Jürgen Schukraft joined CERN's heavy-ion programme in 1986 and was the first spokesperson of the ALICE experiment at the Large Hadron Collider

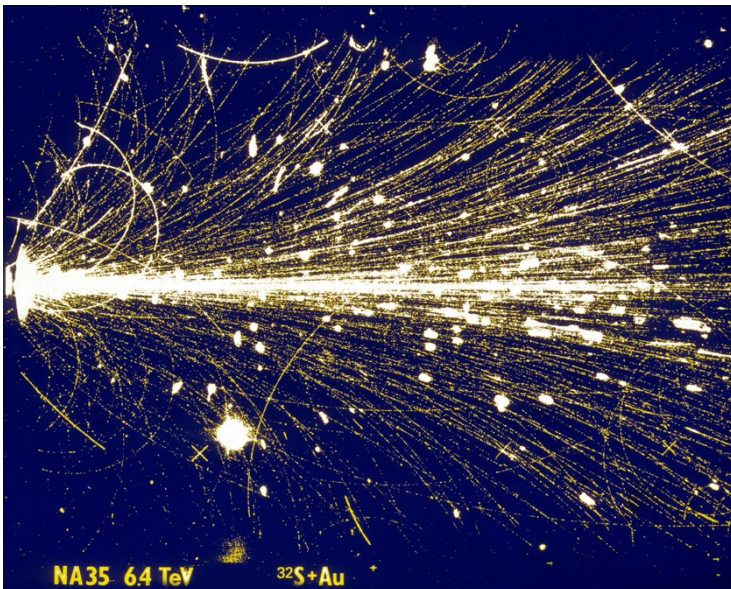


The ALICE detector in 2019, at the start of its [upgrades](#). (Image: CERN)

On 7 November 2010, lead nuclei collided in the [Large Hadron Collider \(LHC\)](#) for the first time. In the control room of ALICE, the LHC's heavy-ion experiment, a celebratory atmosphere reigned. This was the beginning of a new chapter in CERN's heavy-ion programme, which had started 25 years earlier.

In the 1980s, theoretical work had indicated that high-energy heavy-ion collisions – using, for example, lead or gold nuclei – would allow quark–gluon plasma, a state of matter thought to have existed at the very start of the Universe, to be reproduced. Less than 10 microseconds after the Big Bang, the Universe was too hot and dense for quarks and gluons to bond together to form protons and neutrons, the constituents of atomic nuclei as we know them. Quarks and gluons moved around freely in a sort of primordial soup known as [quark–gluon plasma](#).

In 1986, the programme's first experiments were carried out using relatively light nuclei, such as oxygen and sulphur nuclei, which were fired by the [SPS \(Super Proton Synchrotron\)](#) accelerator at targets made of heavier elements, such as gold and lead. These experiments collected promising clues, but did not produce conclusive results confirming the existence of the much-talked-about primordial soup. A second programme started in 1994 with lead ion beams. A total of seven large experiments (and other smaller ones) measured different aspects of the collisions. The combined results collected by these experiments allowed CERN to announce the discovery of a "[new state of matter](#)" in February 2000. At the same time, the Brookhaven National Laboratory in the United States started up its Relativistic Heavy Ion Collider (RHIC). In 2005, the Brookhaven Laboratory announced that it had observed quark–gluon plasma and its characteristics were unexpected. The plasma behaved like an [almost perfect liquid](#) with practically no viscosity.



An event display showing a collision between a beam of sulphur ions and a target containing gold atoms, recorded in 1991 by the NA35 experiment, one of the first experiments in CERN's heavy ion programme. (Image: [CERN](#))

During this time, at CERN, the Large Hadron Collider collaborations were preparing to assemble their detectors. In 1993, a group of scientists led by Jürgen Schukraft had proposed the construction of a detector for the study of heavy-ion collisions, in order to carry on the programme that had started a few years earlier at the SPS. Known as [ALICE \(A Large Ion Collider Experiment\)](#), the experiment was designed to identify the multitude of particles emitted by heavy-ion collisions. With the first proton collisions, which were followed by lead-ion collisions in 2010, ALICE started exploring the strong interaction and quark–gluon plasma.

In 15 years of operation, ALICE, along with the other [LHC experiments](#), has produced a large quantity of detailed results that allow us to better understand the state of matter at the very beginning of the Universe. The experiment has, for example, studied the behaviour of plasma and

determined the characteristics of its flow. It has also observed the phenomenon of jet quenching, which reveals the energy loss of quarks and gluons in plasma, and has defined more precisely the production of the various particles in heavy-ion collisions.

Recollections

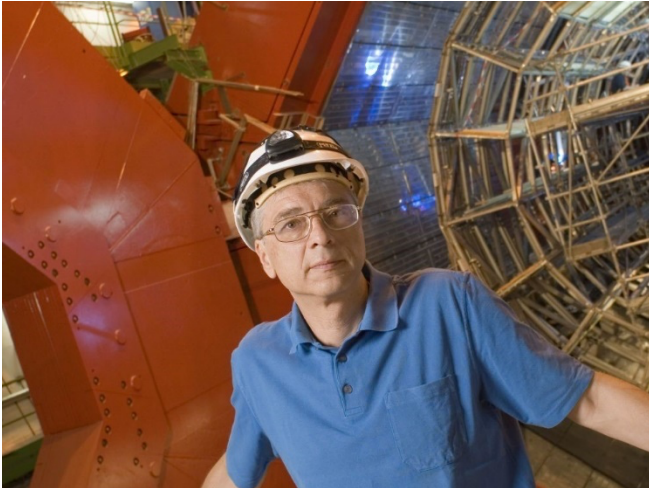
After two years at the [Intersecting Storage Rings \(ISR\)](#), Jürgen Schukraft joined CERN's heavy-ion programme when it started data taking in 1986. He worked on the North Area experiments NA34 and NA45 before starting at ALICE, CERN's heavy-ion experiment at the Large Hadron Collider (LHC), for which he was the first spokesperson from 1993 to 2011.

"The beginning of the heavy-ion programme at CERN in the 1980s was both exciting and confusing. We made a big leap in energy with SPS and thought that it might be enough to create a quark–gluon plasma (QGP). Everyone was in a "gold rush" mood, but nobody knew what we

would find or when. During the programme's first days, Austrian TV came to CERN wanting to film the "first detection of the quark–gluon plasma" live. They were disappointed when they realised that it would take more time...

Even though we found two striking signatures that were predicted for a QGP, the first results were confusing. Better theory was needed, and more data. From 1994, we could use heavy ions (lead), which made all the difference. Plus, a new generation of bespoke detectors opened new possibilities, and evidence that we were looking at something "different" was accumulating. In early 2000, [CERN announced "compelling evidence" for](#)

[a new state of matter](#). Yet, we still couldn't be sure if this "new state" was the real QGP or only a precursor, or even an imposter.



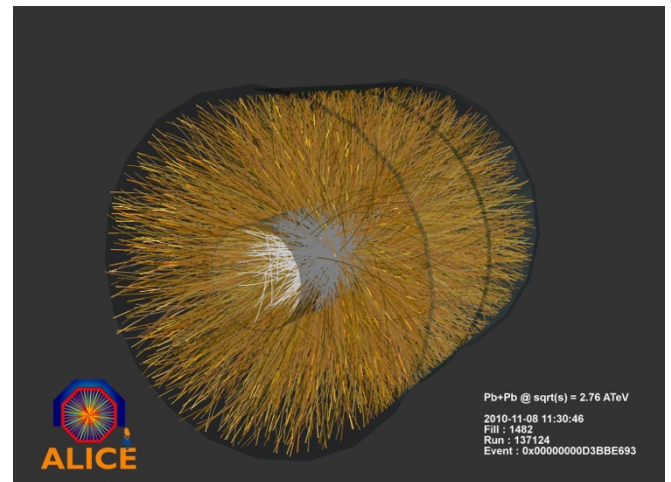
Jürgen Schukraft in 2006 in front of the ALICE experiment being assembled. (Image: [CERN](#))

The centre of action moved to Brookhaven National Lab with the startup of RHIC in summer 2000. By 2005, BNL declared that yet another new state of matter had been found – a "[strongly interacting perfect liquid](#)" – an even more fluid substance (with less internal friction) than superfluid helium. Around the same time, the SPS confirmed another crucial prediction, the "melting" of hadron masses in the QGP and also precisely measured the QGP temperature, far above the predicted threshold. By then we knew, in hindsight, that we had been dealing with the QGP all along.

Back at CERN, we were busy inventing new detector technologies for the LHC, building and assembling the (for us) gigantic ALICE experiment, and writing thousands of pages of technical documents. After [an aborted attempt the year before](#), the LHC saw "first light" during a short proton-proton test run in November 2009. The packed ALICE control room erupted in applause when the first collision "event" popped up on the screen. The display looked good, the detectors worked and decades of preparation had come to a successful end.

A young fellow sitting quietly in the back of the room had single-handedly reconstructed and analysed the 284 (!) events that we netted during the test. Within an hour, he interrupted our celebration with the first actual physics result, the density of produced particles. It looked good, and the next morning, literally in the shower, I had the

crazy idea to see if we could push this all the way to publication. The following days were the most intense and exhilarating of my life. The detector had worked out of the box, but of course not perfectly; in intense meetings twice a day, with a 24-hour task force of a few dozen people, we sorted out one problem after another. By day five, one minute after midnight, the paper was submitted – a full week before the first scheduled LHC collisions. It took 20 years to build the ALICE detector, 40 minutes to take the first data, one hour for preliminary analysis, two days for the (almost) final results, and three days to agree on the author list. The ship had set sail, and the physics journey into ALICE's wonderland had begun in earnest.



One of the first lead-ion collision recorded by ALICE in 2010 (Image: CERN)

And what an expedition it was. In the years since, the LHC heavy-ion programme has helped to resolve most of the earlier puzzles. Notably, we understood the confusing ups and downs of charmonium suppression as an unanticipated consequence of the QGP. With heavy-ion results also coming in from the other three large LHC experiments, we measured the properties of the QGP with ever-increasing sophistication and precision. And early on, still with proton beams, we tripped over another surprise – the first, and arguably still the most unexpected, LHC discovery. A feeble, but distinct, very long-range correlation between particles, never before seen in any elementary collisions, was announced in 2010 to a packed auditorium by the CMS Spokesperson with an apology ("we present this signal to the scrutiny of the scientific community because we didn't

succeed in killing it"). The mysterious result spawned many different explanations, which were mostly ad hoc, speculative, or outright weird. Today, after a plethora of painstaking and detailed measurements, including crucially in proton–nucleus collisions, and alongside great strides in theory, we understand this correlation – unexpectedly appearing in “lowly” proton–proton collisions – as a precursor or onset of the hot and dense matter phenomena we had considered the exclusive territory of heavy-ion reactions.

Forty years after seeing (or rather, understanding) not much in those first light-ion collisions at the SPS, we are now regularly exploring and testing with remarkable detail and precision QCD

(quantum chromodynamics), the theory of the strong interaction, in the regime where it actually is strong. Along the way, we have found traces and shadows of the QGP everywhere we looked, including in the lightest of systems. Such are the surprises and adventures of exploring nature at the frontiers of our understanding, and it has been a privilege – and lots of fun – to be part of it for so long.”

This interview is adapted from the 2004 book “Infinitely CERN”, published to celebrate CERN’s 50th anniversary, updated by Jürgen Schukraft in 2024.

Particle physicists chart a course to the future

The aim of the third update of the European Strategy for Particle Physics is to develop a visionary and concrete plan that greatly advances human knowledge in fundamental physics through the realisation of the next flagship project at CERN

Geneva, 11 October 2024. During its two hundred-and-eighteenth session, the CERN Council formally finalised the organisation of the European Strategy process by appointing the Physics Preparatory Group, whose aim will be to prepare the scientific contribution based on the input from the community. The European Strategy is driven by the CERN Council, the highest authority of the Organization, composed of representatives of all Member States. During the next year and a half, the whole community will work to develop a common vision for the future of particle physics in Europe – including which major project should succeed the Large Hadron Collider at CERN. The process is expected to be concluded at the end of 2025, after which the European Strategy Group will submit its recommendations to the Council.

Particle physics, the study of the smallest constituents of matter and the laws of nature at the most fundamental level, is at a fascinating juncture. The discovery of the Higgs boson at CERN’s Large Hadron Collider (LHC) in 2012 confirmed the final predicted piece of the Standard Model of particle physics – a powerful quantum theory which describes three of nature’s four fundamental forces. Yet, despite its unparalleled success, the Standard Model explains only 5% of the Universe and is not able to answer

other outstanding questions. It cannot account for the invisible “dark matter” that influences the motion of galaxies, nor what is causing the expansion of the Universe to speed up. It also falls short in explaining why matter and antimatter did not annihilate each other in the first moments of the Big Bang to leave a featureless sea of radiation. Studying the Higgs boson and other elementary particles in greater detail and exploring nature at higher energies using next-generation colliders will be key in tackling these and other profound open questions.

“The discoveries at the LHC of the Higgs particle and more than 70 new composite hadrons have been highlights of an amazing decade in particle physics dominated by experiments at CERN, which will be continued at the LHC and its high-luminosity upgrade until 2041. Researchers across the globe agree that deeper study of the Higgs particle is certain to lead to very valuable scientific results, and the CERN Council is united in its vision to have CERN continue to provide the most exciting experimental scientific programme in high energy physics with the aid of the best possible technology. The community and the Council therefore eagerly await the recommendations of the European Strategy Group on the best path to

be taken", said CERN Council President, Eliezer Rabinovici, Hebrew University of Jerusalem.

The third update of the European Strategy for Particle Physics was launched by the CERN Council on 21 March. In June, the Council elected Professor Karl Jakobs (University of Freiburg) as Strategy Secretary and established the European Strategy Group, which will be responsible for submitting final recommendations to the Council for approval in early 2026. The European Strategy Group remit states that the aim of the Strategy is to develop "a visionary and concrete plan that greatly advances human knowledge in fundamental physics through the realisation of the next flagship project at CERN".

The previous update of the European Strategy for Particle Physics, completed in 2020, recommended that Europe, together with its international partners, should investigate the technical and financial feasibility of a future hadron collider at CERN with a centre-of-mass energy of at least 100 TeV and with an electron-positron Higgs and electroweak factory as a possible first stage.

Since then significant progress has been made and an international consensus reached on the scientific case for a Higgs factory. A mid-term report on the feasibility study for the proposed multi-stage Future Circular Collider (FCC) at CERN was presented in March 2024, with a final report expected in spring 2025. In December 2023, a rigorous planning exercise in the US prioritised support for a Higgs factory based outside the US. It was followed in April 2024 by a joint statement of intent by CERN and the US government to continue collaboration on the feasibility study for the FCC Higgs factory (FCC-ee) and on its construction and physics exploitation, should the CERN Member States determine the FCC-ee to be CERN's next world-leading research facility. "Given the long timescales involved in building large colliders, it is vital that the community is united so as to enable the Council to take a decision on the next collider at CERN in 2027/2028. There is excellent progress with the LHC and no new indication that would change our physics priorities: understanding the Higgs boson much better and exploring further the energy frontier are key to the next project", said Strategy Secretary, Karl Jakobs, University of Freiburg.

In addition to identifying the preferred option for the next collider at CERN, the Strategy update is expected to prioritise alternative options to be pursued if the chosen preferred plan turns out not to be feasible or competitive. It will also indicate areas of priority for scientific exploration complementary to colliders, as well as for other items identified as relevant to the field. These include accelerator, detector and computing research and development, theory developments, actions to minimise environmental impact and improve the sustainability of accelerator-based particle physics, initiatives to attract, train and retain early-career researchers, and public engagement.

During its September session the Council appointed members of the Physics Preparatory Group, which will prepare scientific input to the work of the European Strategy Group based on the views it gathers from the community. The Council also announced that the Strategy Open Symposium, at which researchers will be invited to debate the future orientation of European particle physics, will take place in Venice from 23 to 27 June 2025.

"This year we celebrate seven decades of groundbreaking progress in fundamental science and technology at CERN, achieved through international collaboration. This success is the result of CERN's unique culture and the CERN community's boldness in pursuing projects at the limit of human capabilities and has only been possible thanks to the support of our Member and Associate Member States and our partners from all over the world. Future projects will be even more challenging, but their expected physics and technological impact is immense. The future of CERN and the field is bright", said CERN Director-General, Fabiola Gianotti.

About the European Strategy for Particle Physics

The European Strategy for Particle Physics is an open, inclusive and evidence-driven process and takes into account the worldwide particle physics landscape and developments in related fields. It was initiated by the CERN Council in 2005, when completing the Large Hadron Collider (LHC) was named as the top scientific priority. The first strategy update, adopted in 2013, continued to prioritise the LHC and its high-luminosity upgrade

and stated that Europe needed to be in a position to propose an ambitious post-LHC accelerator project at CERN by the time of the next update. The second update, completed in 2020, recommended an electron-positron “Higgs factory” collider as the highest-priority next facility and, in the longer term, that Europe should have the ambition to operate a proton-proton collider at the highest achievable energies.

The third strategy update, launched on 21 March 2024, invites the community to submit written input by 31 March 2025. This will be followed by a scientific open symposium in Venice from 23 to 27 June 2025, where the community will be invited to debate the future orientation of European particle physics. A “briefing book” based on the input and

discussions will then be prepared by the Physics Preparatory Group. The briefing book will be submitted to the European Strategy Group by the end of September 2025 for consideration during a five-day-long drafting session which is scheduled to take place from 1 to 5 December 2025. The European Strategy Group is expected to submit the proposed strategy update to the CERN Council by the end of January 2026.

Further information:

Strategy webpage:

europeanstrategyupdate.web.cern.ch/welcome

2020 Strategy update:

<https://cds.cern.ch/record/2720129>

Estonian flag raised at CERN

CERN raised today the Estonian flag to welcome its 24th Member State



Flag raising ceremony in presence of Vice-President of the CERN Council, Eric Laenen, President of the Republic of Estonia, Alar Karis, CERN Director General, Fabiola Gianotti and the Ambassador of Estonia, Riia Salsa-Audiffren (from left to right). (Image: CERN)

The Estonian flag was raised today at a ceremony held at CERN to mark [the country's accession as CERN's 24th Member State](#). The ceremony was attended by the President of the Republic of Estonia, Alar Karis, one of the two Vice-Presidents of the CERN Council, Eric Laenen, and the CERN

Director-General, Fabiola Gianotti, together with representatives of CERN's Member and Associate Member States and Observers and the CERN community.

“I am delighted that the aspirations of our academic community, companies and government have come together and made CERN membership a reality. This is a major milestone for Estonia and we are keen to contribute to the exciting research and innovation at CERN,” said Alar Karis.

“On behalf of the CERN Council, I am very pleased to welcome Estonia as a Member State, building on decades of commitment to basic science and peaceful scientific collaboration. We are now looking forward to reinforcing Estonia's participation in the CERN Council,” said Eric Laenen.

“The raising of the Estonian flag is a highly symbolic culmination of Estonia's journey to becoming a CERN Member State, which recognises the long-standing fruitful scientific cooperation between Estonia and CERN and Estonia's significant contribution to fundamental research. We look forward to enhancing our collaboration in particle physics research, technology and innovation and in the training and education of the next generations with Estonia as a Member State,” said Fabiola Gianotti.

CERN teams up with Luxembourg and the World Food Programme to tackle global hunger

The new partnership aims to leverage state-of-the-art AI technology to speed up progress towards a world with zero hunger



Representatives from WFP, LIST and CERN met in Luxembourg to sign a new strategic partnership that aims to reduce world hunger. (Image: MAE Luxembourg)

A new partnership funded by the Government of Luxembourg, the [United Nations World Food Programme](#) (WFP), the [Luxembourg Institute of Science and Technology](#) (LIST) and CERN aims to use expertise in AI technology to address the issue of global hunger. The two-year strategic partnership will explore three use cases in which CERN's AI algorithms for weather modelling and prediction, as well as its expertise in distributed infrastructures, will be used together with LIST's AI for Earth observation and crop modelling. These technologies will be harnessed to amplify WFP's impact by increasing the efficiency and effectiveness of its emergency response and development activities.

The partnership is funded by the Government of Luxembourg as part of the country's development cooperation strategy and humanitarian emergency relief efforts. "In the face of multiple intersecting emergencies and disasters that are plunging the world into a deepening global hunger crisis, rethinking our approaches to dealing with humanitarian crises has become a necessity. I am proud that Luxembourg's expertise, knowledge and leadership in promoting AI can make a significant contribution to this unique partnership, which aims to foster cutting-edge technologies to accelerate progress toward a world without hunger and enhance humanitarian action worldwide", says Xavier Bettel, Deputy Prime Minister and Minister for Development Cooperation and Humanitarian Affairs, Luxembourg.

WFP helps food-insecure communities to prepare for, respond to and recover from climate shocks and stresses. It also implements climate and civil infrastructure risk management solutions in 42 countries, benefitting more than 15 million people. "This collaboration allows us to leverage responsible AI to transform our field operations, providing country offices with cutting-edge tools to make data-driven decisions in real time," says Rania Dagash-Kamara, Assistant Executive Director for Partnerships and Innovation at WFP. "By integrating Luxembourg's commitment to innovation, CERN's technological expertise and LIST's Earth observation capabilities, we are setting a new standard for humanitarian response that can save lives and enhance food security for communities in crisis."

The first use case for WFP's partnership with CERN and Luxembourg proposes to extend the [WFP SKAI platform](#), a tool developed with Google Research that uses AI and satellite imagery to enable real-time insights and actionable intelligence for effective decision making during disaster response. The goal is to enhance the platform with improved object detection, support for more data types, and better reliability, scalability and performance and to integrate short- and medium-term weather modelling and forecasting capabilities. The existing and new technologies being developed by CERN, LIST and WFP will be combined to form a "multimodal digital twin" proof-of-concept. This will use image and sensor data with AI-based simulations, moving towards decision support systems that use data from different domains. "LIST is honoured to take part in this strategic collaboration with the World Food Programme and CERN in AI for Earth observation and crop modelling to support humanitarian aid agencies", says Lucien Hoffmann, Science Director at LIST. "LIST brings its recognised expertise in integrating Earth observation data with in-situ measured data, land surface models and satellite and terrestrial communication services in order to provide evidence-based decision support tools in

near real time for one of humanity's greatest challenges related to food security."

A second use case proposes to adapt the CERN multiformat anomaly detection algorithms used for physics data monitoring and analysis to the WFP cash-based transfer system. Cash-based transfers constitute 35 percent of all WFP assistance, making WFP the world's largest provider of humanitarian cash. WFP sends money to people in need to save them from having to make impossible trade-offs, like deciding to eat less so that they can keep their children in school. The CERN anomaly detection algorithms could provide a tool for the early detection of cash transfer misuse.

To support secure and reliable data processing in the first two use cases, a third use case proposes to adapt the CERN federated learning platform (currently used in healthcare) to enable the training of AI models using data from multiple

organisations for a broad range of applications. These will be defined by WFP and other organisations currently looking for secure ways of processing sensitive data.

"CERN has a long tradition of excellence and over the years has developed strong AI capabilities and competencies to enable its cutting-edge scientific research programme," concludes Enrica Porcari, Head of the IT department at CERN. "This partnership confirms once more CERN's role in fostering collaborations that address common challenges and contribute to transfer knowledge from science to society to create a positive impact for future generations."

Read more:

- [CERN and the World Food Programme sign a memorandum of cooperation](#)

Antonella Del Rosso

Decoding top quarks with precision

The ATLAS experiment at the LHC reveals how pairs of top quarks are produced along with other heavy quarks

In collisions between protons at the Large Hadron Collider ([LHC](#)), pairs of top quarks – the heaviest known elementary particles – are frequently produced along with other heavy quarks, including bottom and charm quarks. These collision events can provide physicists with valuable insights into quantum chromodynamics (QCD), the theory that describes the strong force. Precisely determining the production rates (or "cross-sections") of these processes also enables researchers to more effectively distinguish them from rarer phenomena.

In two recent studies, the [ATLAS](#) collaboration analysed proton–proton collision data from LHC Run 2 (2015–2018) to measure how often top-quark pairs are produced alongside bottom quarks or charm quarks and to explore the detailed dynamics of these processes.

The [first ATLAS study](#) focused on how often top-quark pairs are created along with "jets" of particles produced by bottom quarks (b-jets). To identify (or "tag") these b-jets with high accuracy, researchers used tools called flavour-tagging

algorithms, which look for specific patterns, such as how far through the ATLAS detector particles traverse before they decay.

By analysing events with opposite-charge electron–muon pairs and at least three or four b-jets, ATLAS physicists made the most precise measurements to date of the total cross-sections of this top-quark-pair process in these event categories. These measurements surpass the accuracy of current theoretical predictions, especially in events with higher numbers of b-jets. The results were compared to multiple theoretical predictions to assess the accuracy of the modelling of this process.

The researchers also examined motional (or "kinematic") properties, such as the momenta of the b-jets and the total jet energy, to test how well simulations of top-quark-pair production match the data. Their findings point to areas where theoretical models need improvement to better capture the complexities of additional b-jet production.

The [second ATLAS study](#), presented recently at the [17th International Workshop on Top Quark Physics](#), broke new ground by providing the first dedicated ATLAS measurement of how often top-quark pairs are produced along with jets originating from charm quarks (c-jets). ATLAS physicists analysed events with one or two leptons (electrons and muons), using a custom flavour-tagging algorithm developed specifically for this study to distinguish c-jets from b-jets and other jets. This algorithm was essential because c-jets are even more challenging to identify than b-jets, as they have shorter lifetimes and produce less distinct signatures in the ATLAS detector.

The study found that most theoretical models provided reasonable agreement with the data, though they generally underpredicted the

production rates of c-jets. These results, which for the first time separately determined the cross-sections for single and multiple charm-quark production in top-quark-pair events, highlight the need for refined simulations of these processes to improve future measurements.

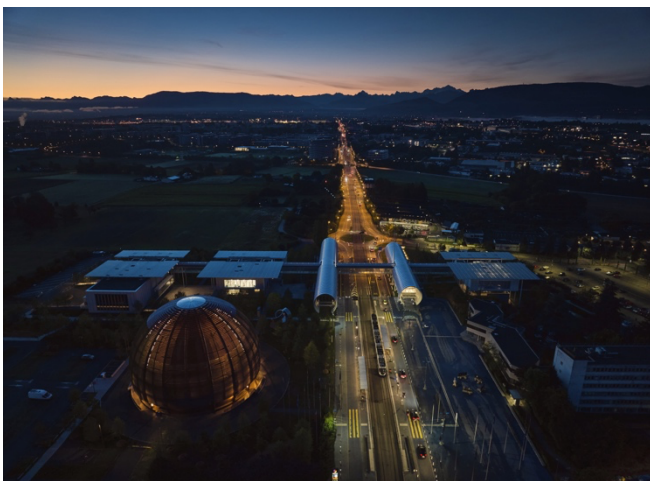
Taken together, these studies deepen the understanding of the top quark's relationship with bottom and charm quarks within the framework of QCD. In addition, they pave the way for new explorations of rarer processes involving top quarks – such as the simultaneous production of [four top quarks](#) – that could push the boundaries of known physics.

Read more on the [ATLAS website](#).

ATLAS collaboration

How can I use CERN Science Gateway?

From shop discounts to team-building lab workshops and more: find out how CERN's one-year-old education and outreach centre can benefit the CERN community



CERN Science Gateway at sunrise. (Image: CERN)

What does CERN Science Gateway offer?

Science Gateway is CERN's flagship centre for education and outreach. Since opening on 7 October 2023, it has welcomed almost 390 000 visitors from 171 different countries to discover CERN's scientific goals, research and discoveries, as well as its technologies, becoming one of the most visited places in Switzerland. The centre offers exhibitions, science shows, a multitude of lab workshops, a café and a shop. It also boasts a large space for events for the CERN community, as well as for public and private events, which has hosted 80 events in the last year.

[Featured in TIME magazine's "World's Greatest Places"](#), it's a great place to take your friends and family, to show them what you do. But there's even more on offer for the CERN community, too.

Can I get discounts in the CERN shop?

Yes! [Anyone](#) with a CERN badge can get 10% off at [the shop](#), all year round. From physics-themed T-shirts to a miniature LHC Lego set, you can find gifts for you, your friends and your family.

Can I arrange team-building lab workshops?

Yes – workshops are great for a team-building activity with your colleagues or a visit from family, friends or collaborators who would like some hands-on science experience.

Two workshops, "[Seeing the Invisible](#)" and "[Cloud Chamber](#)" are available for visits organised by members of the CERN personnel.

You can request to book a lab workshop using the timeslots that are not taken up for standard workshops. You will need **your own CERN official guide** who can run the workshop. Please read the information [here](#) carefully before submitting a request.

What rooms are available and how can I book them?

The Science Gateway Auditorium and the Globe of Science and Innovation are available for events (co-)organised by CERN, as well as private events. There are multiple ways to configure these spaces depending on the size of your event. Please be aware that the Science Gateway campus is not entirely funded by CERN, so booking the space will require a CERN budget code and a supporting email from the head of the requesting department. Visit [this webpage](#) for more details on the conditions of use and to place a booking request.

How can I be more involved with CERN Science Gateway?

The best way is by becoming a CERN guide. To do this, you can join the [introduction course](#) held

every 3–4 weeks, which will give you an overview of what being a CERN guide entails. From there, further courses are available depending on what type of guide you would like to be.

For more questions, you can check the [guides website](#) or contact guides.manager@cern.ch. Remember, becoming a guide can [benefit you](#), too.

How can I find out more about Science Gateway?

CERN Science Gateway is open from 9 a.m. to 5 p.m., Tuesday to Sunday, including bank holidays (with some [exceptional](#) closures). You can explore the centre yourself, or the team is available to answer your questions on site. Visit the [website](#) for more information.

We look forward to seeing you there!

Work Well Feel Well: Adopting the strategy of an elite athlete in the workplace

Part seven of the [Work Well Feel Well](#) series looks at how to prepare for peaks in workload



Preparing physically and mentally

The workload, fast pace and pressures of our professional lives can require performance levels similar to those of a competitive athlete.

While challenges can be stimulating and motivating, as time goes on they can also lead to the risk of exhaustion.

The strategy of an elite sportsperson can show us how to best manage high-level pressures on our body and our mind. Athletes manage their training rhythm and maintain their efforts throughout the year in cycles of three phases, organising each cycle around a competition:

1. **Preparation:** athletes need to train physically and prepare mentally before a competition. This can involve using new equipment and practising different techniques.
2. **Maximum pressure:** during a competition, athletes need to be fully focused. They must conserve their resources carefully, get a good night's sleep and eat healthily.
3. **Active recovery:** as the pressure eases after a competition, light training and stretching – as well as reflecting on possible changes for the future – can help an athlete to recover and prepare for the next big event.

In the workplace, our challenge is to develop work endurance and maintain our energy levels over the long term. Just like an athlete, we need to manage our rhythm and efforts to prepare for our own “competitions” – the periods of the year with the greatest workload.

What preparations do we need to manage stress during times of pressure?

After an intense and heavy workload, what can we do to relieve the pressure and actively recover?



On your marks: participants at the start of the 2024 CERN Relay Race. (Image: CERN)

Take action

As part of the “Efficiency and caring at work” campaign, the [Work Well Feel Well website](#) now offers useful resources that can be downloaded, including an exercise on respecting your own rhythm.

Other suggestions include:

- **Stroke awareness:** physical activity, a balanced diet and reduced stress are some of the ways to reduce the risk of strokes and cardiovascular diseases. Find out more [on 17 October](#), when the CERN Medical Service and local partners are running a [stroke awareness campaign](#) ahead of World Stroke Day on 29 October.
- **Cardiac coherence:** watch the recording of the talk by the CERN Medical Service and psychologists on the topic of “[cardiac coherence](#)”, during which they offer [practical advice and suggest exercises](#) to help you to take a moment to find your own rhythm.
- **Training courses:** the [CERN training catalogue](#) includes courses on how to [work smarter not harder](#), [balance performance and pressure](#) and [assess resilience](#), to help you to look after yourself before, during and after periods of intense work.

HR department

CERN-Solvay student camp: meet the next generation of scientists

The latest camp just concluded – 30 high-school students from all over the world came to CERN for a week full of science and fun

[Two years ago](#), CERN and [Solvay](#), a Belgian science company, launched [the CERN-Solvay Education Programme](#), a three-year partnership to inspire high-school students from around the world to get involved in science, technology, engineering and mathematics (STEM) subjects and aspire to related careers.

The programme combines remote and onsite learning. It begins by sparking the curiosity of students on social media with a collection of [short videos linking homemade experiments](#) to CERN physics and technologies. Interested participants can then dive deeper by enrolling in [an online course](#) to understand CERN physics and technologies. Lastly, the students who pass the online course can apply to the [CERN-Solvay student camps](#) held at CERN. For an entire week,

30 high-school students from all over the world have the opportunity to do hands-on experiments, visit research facilities at CERN, attend lectures and Q&A sessions, explore STEM careers and take part in social activities. [The first camp](#) took place in October 2023, the second one in May 2024 and the latest one has just concluded.

“Seeing how engaged these students were was inspiring,” [wrote](#) Solvay CEO Philippe Kehren, who spent a day with the students at the latest camp. “They asked interesting questions about the effects of science on society and the environment. This exchange of ideas shows how we help future innovators. We provide them with the insights they need to tackle important global issues.”



The latest CERN-Solvay student camp took place at CERN from 6–12 October 2024. [Click here](#) to find out what CERN and Solvay had in store for the students. (Image: CERN)

“Last week's camp was truly a highlight of the CERN-Solvay Education Programme,” said Guillaume Durey, the programme manager at CERN. “Years of hard work culminated in an event that surpassed the high standards set by the previous two camps. It was incredibly rewarding to witness how deeply moved the participants were by everything they experienced at CERN. We’re looking forward to guiding many more students towards bright futures in physics and beyond.”

Anaïs Schaeffer

The CERN-Solvay Education Programme is part of the education portfolio of [Science Gateway](#), CERN's flagship centre for education and outreach. Since opening on 7 October 2023, CERN Science Gateway has welcomed almost 390 000 visitors from 171 different countries to discover CERN's scientific goals, research and discoveries, as well as its technologies, becoming one of the most visited places in Switzerland. Members of the CERN community can find out more about what Science Gateway can offer them in [this article](#).

Computer Security: Bug Bounty begins

“@CERN #BugBounty Program launches today! Did you know that the WWW as we know it today was created at CERN? Thanks to this fundamental invention, you now have the opportunity to help @CERN Computing uncover vulnerabilities. With a maximum bounty of 2^{10} CHF rewarded to those who spot and present groundbreaking findings. Join the program to make a cosmic impact and illuminate the black holes of CERN's digital realm. #Cybersecurity”. This is the announcement that the CERN-chosen security company has published to its hacker community to invite them to start hacking CERN.

After an all-encompassing computer security scan for vulnerabilities in CERN's web sphere by another external company, we have launched the next stage: the Bug Bounty. While the security scan was intended to be broad – targeting about 2500 web servers with openings in CERN's outer perimeter firewall towards the internet and nearly

7500 websites that are open and unprotected by CERN's SSO – the Bug Bounty will poke deep. While the automatic scan produced more than 35k unverified findings, where the majority were (wrongly!) linked to outdated versions (but with backported security fixes!), as well as several dozen more critical problems like (mainly!) cross-site scripting and a few not-so-problematic command line injection vulnerabilities, the Bug Bounty will discover, verify and validate more complicated exploits that could potentially lead to a takeover of CERN's IT infrastructure as a whole. For instance, breaking into websites in order to obtain access to the underlying server's operating system, manipulating a badly protected database in order to access confidential information or (possibly with the help of the former two!) compromising web servers in order to get a stronghold in CERN's internal network and then

move laterally through it for more juicy targets (like CERN's AD/LDAP servers).

In parallel, the autumn will also see numerous students from, for the first time, the Deggendorf Institute of Technology in Germany and, once again, the St. Pölten University of Applied Sciences in Austria, who will probe CERN's web sphere as part of [CERN's WhiteHat Challenge](#). In the past, this challenge has already found a multitude of sophisticated vulnerabilities and misconfigurations.

As stated in the announcement, the Bug Bounty programme comes (as its name suggests!) with a cost to CERN in order to create the necessary incentives for hackers to participate. Scaling with the severity of their findings (as rated by the "Common Vulnerability Scoring System" CVSS), the Bug Bounty hackers are paid 2-to-the-power-of-the-finding's-CVSS-score (2^{CVSS}) Swiss francs for each of their findings (so up to 1024 CHF for boss-

level findings – and probably more in the future). However (lucky you!), contrary to what was said in [our controversial Bulletin article on this subject earlier this year](#), we will not yet cross-charge the costs to the owner of the vulnerable website. Nevertheless, we will politely inform them of the corresponding costs in order to give them an idea of the price tag that negligence, lack of training or pure ignorance entails for CERN. And this is just the minimum price tag; a malicious attacker will not ask for a bounty but will instead dig deeper up to the damage level, to either sabotage CERN, exfiltrate data or [ransom CERN](#) to demand much more money than a Bug Bounty costs. So, let's keep CERN's costs at bay. [Keep your websites secure!](#)

Computer Security Office

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

17 October: Annual Information Meeting of the Pension Fund

Update 18 October 2024: This announcement could not be included in the previous Bulletin due to logistical reasons. We apologise for the inconvenience.

However, you can watch the recording of the webcast of the meeting, which will be available shortly on the [Indico event page](#).

All members and beneficiaries of the Pension Fund are invited to attend the Annual Information Meeting, which will take place:

**on Thursday, 17 October 2024
from 2.30 p.m. to 3.30 p.m.**

**in the Council Chamber (503-1-001)
and by [webcast](#)**

French/English interpretation will be available via Zoom videoconference.

As well as providing an update on the Pension Fund, the PFGB Chair and the Fund's Chief Executive Officer will be pleased to answer any questions you may have. Members and Beneficiaries present in the Council Chamber or connected via Zoom videoconference will have the possibility to submit questions.

The 2023 Pension Fund Annual Report and Financial Statements are available on the Pension Fund's website: <https://pensionfund.cern.ch/en>.

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

18 Oct 14:00 | Knowledge Sharing | Main Auditorium & online | Commemorations | EN
[Symposium to celebrate Carlo Rubbia's 90th birthday](#)

18 Oct 16:00 | Knowledge Sharing | BE Auditorium | BE Seminars | EN
[PUMA Experiments](#)

23 Oct 14:00 | Physics | Council Chamber | Theory Colloquia | EN
[The Higgs Boson: From Theory to Experiment](#)

25 Oct 14:30 | Knowledge Sharing | Online | CERN Alumni | EN
[Moving Out of Academia to Aviation & Aerospace](#)

31 Oct 19:00 | Knowledge Sharing | CERN Science Gateway | Public event | FR (simultaneous interpreting in EN)
[Dark Matter Day 2024](#)

4–15 Nov | Knowledge Sharing | Main Building Exhibition | CERN & Society Foundation | EN
[CERN & Society Foundation: 10 years bridging science and society](#)

4 Nov 13:00 | Knowledge Sharing | CERN & online | CERN Knowledge Transfer group | EN
[CERN Venture Connect Summit 2024](#)

6 Nov 10:00 | At CERN | Online | Human Resources | EN
[MERIT 2025: Public Meeting in English](#)

8 Nov 9:00 | At CERN | Main Auditorium & online | HSE unit | EN
[Environment Town Hall](#)

8 Nov 11:00 | Knowledge Sharing | The Globe of Science and Innovation | IdeaSquare | EN
[IdeaSquare 10th anniversary](#)

8 Nov 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN
[L&D Micro-talk 13 – Timeboxing](#)

18 Nov 14:00 | At CERN | Online | Human Resources | FR
[MERIT 2025: Réunion publique en français](#)

Registration now open

11–15 Nov | Physics | 4/3-006 | TH Institute | EN
[Light ion collisions at the LHC](#) *bringing together theorists, experimentalists and accelerator physicists*

14 Nov | Knowledge Sharing | CERN | Education locale | FR
[Futur en tous genres](#)

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN
[DRD1 Gaseous Detectors School](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today, this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Save the date: environmental Town Hall event – 8 November 2024

Join us as we explore CERN's current environmental initiatives, its progress towards achieving them and its future goals looking towards 2030



CERN is committed to minimising its environmental impact through a wide range of projects aimed notably at managing energy use, cutting greenhouse gas emissions, conserving water, protecting biodiversity and managing waste. This Town Hall event will provide an overview of the key initiatives under way and highlight the progress made towards meeting CERN's current environmental targets at the end of Run 3, with an update regarding CERN's [third Environment Report \(2021–2022\)](#), published in December 2023. You'll also hear about the updated goals for Horizon 2030 and the specific projects designed to achieve them.

This Town Hall event is not just an update – it's an invitation for you to engage in shaping CERN's

environmental priorities. We'll discuss how each of us can contribute to making CERN's operations more sustainable and to meeting the Organization's commitment to environmental responsibility.

We encourage everyone to attend and be part of this important exchange. Your participation is key to our collective success in fostering a sustainable future at CERN. Don't miss this opportunity to be involved!

When? 9 a.m. to 10.40 a.m., with coffee served at the end

Where? Main Auditorium (500/1-001).

The event will be webcast and a recording will be made available after the event.

Full programme:

<https://indico.cern.ch/event/1456577>

*Benoît Delille, Chair of the CERN Environmental Protection Steering board (CEPS),
and Fabiola Gianotti, CERN Director-General*

CERN70: been there, done that, got the T-shirt (and more...)?

CERN community: you can now order an exclusive T-shirt to be delivered via internal mail or buy limited edition CERN70 items from the CERN shop

Following the sellout success of the T-shirts at the [CERN70 Community Event](#) and the overwhelming demand since then, the CERN70 team is releasing a second – and final – edition of the [official T-shirt](#).

This limited-edition collector's item is available for 8 CHF. Adult T-shirts are now being offered in five sizes (S, M, L, XL and XXL) in both black and white, and children's T-shirts are available in pale blue in two sizes (3–4 years and 7–8 years). Please note

that no additional stock will be produced beyond what is ordered through [this form](#).

Orders must be placed by 31 October at the latest, and payment must be completed for the order to be confirmed. To secure yours, simply fill out the [order form](#) and complete the online payment.

Orders will be delivered exclusively via internal mail a few weeks after the order deadline. For CERN community members without an internal address, please make arrangements for a person

or service at CERN to receive the delivery on your behalf.

Thank you once again for your enthusiasm and for taking part in the [Community Event](#) on 17 September; your presence made the celebrations truly extraordinary!

Don't miss: Limited edition CERN70 items available at the CERN shop

The CERN shop at Science Gateway now has a range of CERN70 commemorative items including tote bags and mugs that will allow you to remember this memorable year. Don't forget that your CERN access card entitles you to a 10% discount on the price displayed in the shop.

Thank you for your support, and we look forward to seeing you with your 70th anniversary souvenirs!

Launch of a new Safety training programme for supervisors

Equipping all supervisors to strengthen CERN's Safety culture

This November will see the launch of a new training programme, "Safety Matters for Supervisors". This comprehensive training course aims to enhance the integration of Safety management practices into the Organization's general management practices and thus foster a robust Safety culture at CERN. This initiative is designed to ensure that supervisors at all levels are fully aware of their Safety roles and responsibilities and are equipped with the necessary tools and support.

The programme will cover all facets of Safety at CERN, including occupational health and safety, with a focus on well-being and psycho-social risks, environmental protection and safe operational practices. Participants will receive a comprehensive Safety awareness toolbox, designed to help them develop a strong Safety network and effective management practices, positioning them as active contributors to a sound Safety culture.

Following the success of pilot sessions held in May and June 2024, the regular sessions will commence in November, starting with four sessions per year with the possibility of adding more, if needed. The training will be delivered through a blended format, beginning with three e-learning modules covering key Safety concepts,

followed by two interactive workshops that will focus on practical applications and peer-to-peer learning.

The programme is highly recommended for new supervisors in their first year of appointment. Existing supervisors across all levels in both technical and non-technical fields are also encouraged to enrol when places are available. The programme has been aligned with the HR Learning & Development Leadership learning portfolio and approved by the CERN Learning Board, reinforcing the Organization's commitment to embedding Safety into its core management practices.

Benoît Delille, Head of the HSE unit, sums it up: "This is an excellent opportunity for supervisors to enhance their leadership capabilities and contribute to a safer work environment. We encourage all eligible supervisors to participate in this programme and to take an active role in promoting a strong Safety culture at CERN."

For more information and to sign up, follow [this link](#).

Take part and help to make CERN a safer place for everyone!

HSE unit

Data privacy – think before you throw!

Throwing paper documents containing personal data in the waste bin or the recycling can have serious implications for the person concerned, should the information fall into the wrong hands.

How can I safely destroy paper documents containing personal data?

Use a shredder. If you are unsure of where you can find one, contact your departmental secretariat for information.

Furthermore, do not hesitate to spread the word by [downloading the poster](#) prepared by the Data Privacy Coordination Committee.

So, don't let personal data leak... please use a **SHREDDER**.

Data Privacy Coordination Committee

More information on data privacy at CERN can be found on the [ODP website](#) or obtained from your [departmental data privacy coordinator](#).

Central heating turned on at Meyrin and Prévessin sites

Central heating will be progressively turned on throughout CERN from Monday 7 October 2024. All buildings will be heated within the next few days.

[Delaying turning on the heating](#) on campus was one of the measures of the Energy Savings campaign launched in 2022 and included in the measures foreseen by [the ISO 50001 certification](#). According to these measures, the heating switch-on is no longer defined on the basis of a regular

calendar but on the weather forecast, and more specifically on the average overnight temperature. Each day saved through postponing the heating start represents significant energy savings for the Organization.

We therefore thank everyone for their understanding and supportive efforts.

SCE department

Protect yourself and others: get a flu vaccination

The influenza season is approaching and with it comes the consideration of getting a flu vaccination to protect yourself and others. The CERN infirmary will vaccinate any person working on the CERN site (MPE, MPA, temporary workers and contractors) provided the person brings their own vaccine and that the relevant medical questionnaire at the infirmary confirms there are no contraindications to vaccination. Both conditions have to be met for the vaccination to be administered. A medical prescription will be given to the patient after the vaccination to allow the reimbursement of the vaccine by the person's health insurance provider.

CERN retirees and family members can obtain the flu vaccination in France or Switzerland from their general practitioner or chemist's: the vaccination is also be reimbursed by the CHIS according to its own rules.

The infirmary is open for vaccination Monday to Friday from 8 a.m. to 12.30 p.m. and from 2 p.m. to 4.30 p.m.

For more information, see:

<https://hse.cern/flu vaccination>.

Medical Service

Obituaries

Karel Šafařík (1953-2024)



Our dear colleague and friend, Karel Šafařík, sadly passed away on Monday 7 October. Karel graduated in theoretical physics in Bratislava in 1976, and worked at JINR in Dubna for more than 11 years, participating in experiments on the phenomenology of particle production at high energies at Dubna and at Serpukhov.

In 1990, Karel joined the Collège de France and the heavy-ion programme at CERN and very soon became one of the most influential scientists in the OMEGA series of heavy-ion experiments at the CERN SPS (WA85, WA94, WA97, NA57). In 2002, Karel was awarded the Slovak Academy of Sciences Prize for his contributions to the observation of the enhancement of the production of multistrange particles in heavy-ion collisions at the SPS, one of the pillars of the CERN announcement of the observation of the quark–gluon plasma in [February 2000](#). In 2013, he was also awarded the medal of the Czech Physical Society.

Karel was one of the founding members of the ALICE collaboration. As early as 1991, he was part of the small group working on the design of a dedicated heavy-ion detector for the LHC, which would then become the ALICE experiment. He had a central role in shaping ALICE, from the definition of physics topics and the detector layout, to the

design of the data format, tracking, storage and analysis. He had a central role in convincing the collaboration to introduce two layers of pixel detectors, in order to reconstruct, amidst the thousands of tracks produced in central Pb–Pb collisions at the LHC, the topologies of charm hadrons decaying only a few tens of microns from the primary vertex. At the time, many people considered this to be impossible in heavy-ion collisions, and yet this has become today one of the pillars of the ALICE physics programme and long-term scientific strategy. He was the ALICE Physics Coordinator for many years leading up to and including the first ALICE data taking.

Over the years, Karel also made multiple contributions to ALICE Upgrade studies and became known as the “wise man” to be consulted for the trickiest questions. He had organised the ALICE Upgrade Week in Prague in 2022. He was remembered with a minute’s silence at the beginning of the first plenary session of the ALICE Upgrade Week in Krakow (which he had originally planned to attend), on the day of his passing.

Karel was a top-class physicist, with a sharp analytical mind, a legendary memory and a seemingly unlimited set of competencies ranging from higher mathematics to formal theory, from detector physics to high-performance computing. At the same time, he was a generous, caring and kind colleague, and over the years has variously supported, helped, mentored and guided a large number of ALICE collaborators.

Karel’s passing leaves a huge void. We shall miss him. Our thoughts go out to his family, friends and close colleagues.

*His friends and colleagues in the ALICE
collaboration*

An obituary will also appear in the [CERN Courier](#).